
Central Valley Regional Water Quality Control Board

23 April 2021

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PERCOLATION CAPACITY PRELIMINARY ENGINEERING REPORT; CITY OF LIVINGSTON; DOMESTIC WASTEWATER TREATMENT FACILITY; MERCED COUNTY

On 1 October 2020, the Central Valley Water Board received a technical report titled *Percolation Capacity Preliminary Engineering Report* (Percolation Report) as required by Provision G.3.d. of Waste Discharge Requirements (WDRs) Order R5-2014-0147. Subsequently, on 12 November 2020 and 11 December 2020 the Central Valley Water Board received revised versions of the Percolation Report. The November 2020 and December 2020 Reports were signed and stamped by Alfonso Manrique (RCE 63673) of Gouveia Engineering, Inc. Consulting Engineers and submitted on behalf of the City of Livingston (Discharger).

The enclosed staff memorandum further discusses the Percolation Report and its revisions. Central Valley Water Board staff has reviewed the technical report and considers it complete. The Discharger can begin discharging up to 2.0 million gallons per day (monthly average) in accordance with WDRs Order R5-2014-0147.

NITRATE CONTROL PROGRAM

On 29 May 2020, the Central Valley Water Board issued the Discharger (CV-SALTS ID 1972) a Priority 1 Notice to Comply under the Nitrate Control Program. The WWTF falls within two Priority Zones: Priority 1 – Turlock Sub-Basin (5-22.03) and Priority 2 – Merced Sub-Basin (5-22.04) according to the Department of Water Resources Bulletin 118 (2003) Map.

The enclosed staff memorandum also discusses the Priority Zone re-designation for the WWTF in more detail. The Central Valley Water Board has re-designated the WWTF to solely the Priority 2 Merced Sub-basin. The Priority 1 Notice to Comply for the Turlock Sub-basin will be rescinded.

SALT CONTROL PROGRAM

On 5 January 2021, the Central Valley Water Board issued the Discharger a Notice to Comply under the Salt Control Program. The Discharger must submit a Notice of Intent

KARL E. LONGLEY ScD, P.E., CHAIR | PATRICK PULUPA, ESQ., EXECUTIVE OFFICER

Central Valley Regional Water Quality Control Board

TO: Scott J. Hatton 
Supervising Water Resource Control Engineer

FROM: Alexander S. Mushegan
Senior Water Resource Control Engineer

Denise Soria
Water Resource Control Engineer

DATE: 23 April 2021

**REVIEW OF PERCOLATION CAPACITY PRELIMINARY ENGINEERING REPORT;
CITY OF LIVINGSTON; DOMESTIC WASTEWATER TREATMENT FACILITY;
MERCED COUNTY**

On 1 October 2020, the Central Valley Water Board received a technical report titled *Percolation Capacity Preliminary Engineering Report* (Percolation Report). Subsequently, on 12 November 2020 and 11 December 2020, the Central Valley Water Board received revised versions of the Percolation Report. The November 2020 and December 2020 Reports were signed and stamped by Alfonso Manrique (RCE 63673) of Gouveia Engineering, Inc. Consulting Engineers and submitted on behalf of the City of Livingston (Discharger).

BACKGROUND INFORMATION

The City of Livingston owns and operates a domestic wastewater treatment facility (WWTF) at 7160 North Gallo Road in Livingston. In October 2010, the Discharger submitted a report titled *City of Livingston Percolation Ponds Reconfiguration* (2010 Report) for the WWTF. In order to provide a buffer between WWTF ponds and the Merced River, the 2010 Report proposed to permanently abandon Ponds 4, 5, and 6 along with reconfiguring Ponds 1, 2, and 3. According to the 2010 Report, the reconfiguration project would result in a total disposal capacity of 2.0 million gallons per day (mgd).

The WWTF is currently regulated by Waste Discharge Requirements (WDRs) Order R5-2014-0147. WDRs Order R5-2014-0147 authorizes a discharge flow limit of 1.18 mgd and an ultimate flow limit of 2.0 mgd once the Discharger implements the reconfiguration project described in the 2010 Report, or alternatively, completes a

project to provide sufficient disposal capacity of 2.0 mgd as required by Provision G.3. In addition, Provision G.3 requires the Discharger to demonstrate that Ponds 4, 5, and 6 have been permanently disconnected from the WWTF. Specially, Provision G.3.d. requires the following submittal by 5 March 2020:

Submit a technical report in the form of an engineering report and certification that demonstrates the evaporation/percolation ponds can reliably dispose of WWTF effluent up to a monthly average dry weather flow of 2.0 mgd for Executive Officer approval. The technical report must also demonstrate that Ponds 4, 5, and 6 have been permanently disconnected from the WWTF.

The Percolation Report was submitted to satisfy Provision G.3.d. of the WDRs. This memorandum provides a summary of Central Valley Water Board staff's review of the Percolation Report and its revisions. This memorandum also addresses the Priority Zone designation under the Nitrate Control Program.

PERCOLATION CAPACITY PRELIMINARY ENGINEERING REPORT

The reconfiguration project as proposed in the 2010 Report was never completed. Instead, the Discharger pursued an alternative project that consisted of only utilizing Ponds 7 and 8 and making the technical demonstration it has sufficient disposal capacity to accommodate 2.0 mgd of wastewater with these ponds.

The WWTF currently consist of a headworks, one oxidation ditch, two secondary clarifiers, two scum ponds, four soil cement sludge drying beds, a soil cement sludge holding pad, and eight percolation ponds (Ponds 1 through 8).

As currently configured, wastewater is discharged directly to Pond 7 and 8. According to the Percolation Report, the Discharger filled Junction Box No. 2 with concrete as well as sealed off the discharge point within Pond 4 to ensure that no wastewater can be discharged into Ponds 4, 5, and 6, satisfying a part of Provision G.3.d of WDRs R5-2014-0157.

The Percolation Report demonstrates that the City can rely entirely on Pond 7 and 8 for the disposal of 2.0 mgd of wastewater and does not plan on using Ponds 1 through 3 in the future. In addition, the Percolation Report states that Ponds 4, 5, and 6 have not been in service since 2008. Staff's review of Self-Monitoring Reports from January 2015 through February 2021 indicate that wastewater has not been discharged to Ponds 1 through 6 for several years.

In 2018, Gouveia Engineering performed a field percolation test in Pond 7 that consisted of installing a 48-inch diameter precast concrete pipe vertically in Pond 7. Treated effluent was discharged into the concrete pipe from May 2018 through September 2018. The average percolation rate observed based on meter readings was 3,490 gpd and eventually stabilized around 2,000 gpd (10.6 inches/hour or 255.6 inches/day).

On 8 September 2020, RMA GeoScience performed subsurface exploration of Ponds 7 and 8 down to a depth of thirteen and fourteen feet below ground surface (bgs),

respectively. Soils found below Ponds 7 and 8 consists of fine to medium grained silty sand in the upper three feet of soil followed by coarse-grained poorly graded sands in the remaining ten to eleven feet bgs.

A field permeability rate of 255.6 inch/day for soils below the WWTF appears appropriate. According to Wastewater Engineering, 3rd Edition by Metcalf and Eddy, medium to coarse sands have permeability rates ranging from 396 inch/day to 3,960 inch/day.

WATER BALANCE

According to the Percolation Report, Pond 7 and 8 have a surface area of 7.5 acres and 9 acres, respectively. The Percolation Report's water balance was calculated using only Pond 7 with a percolation rate of 255.6 inch/day. Based on the water balance, it appears the WWTF has sufficient disposal capacity with only one pond (Pond 7) in service to accommodate 2.0 mgd during a 100-year storm event.

NITRATE CONTROL PROGRAM

On 29 May 2020, the Central Valley Water Board issued the Discharger (CV-SALTS ID 1972) a Priority 1 Notice to Comply under the Nitrate Control Program. On 7 August 2020, we received an email inquiry from Mr. Manrique on behalf of the Discharger requesting clarification on which Priority Zone is applicable to the WWTF. The WWTF falls within two Priority Zones: Priority 1 – Turlock Sub-Basin (5-22.03) and Priority 2 – Merced Sub-Basin (5-22.04) according to the Department of Water Resources Bulletin 118 (2003) Map.

The WWTF components that fall within the Priority 1 Zone are: portions of Ponds 1 through 3 (not in use), Ponds 4 through 6 (not in use), and the soil cement sludge holding pad. The WWTF components that fall within the Priority 2 Zone are: the headworks, oxidation ditch, clarifiers, sludge drying beds, Ponds 7 and 8.

COMMENTS

Based on the information presented in the Percolation Report, the Discharger has demonstrated that it has sufficient disposal capacity to accommodate flows of up to 2.0 mgd with solely relying on Ponds 7 or 8. In addition, the Discharger has demonstrated that Ponds 4, 5, and 6 are permanently disconnected from the WWTF. Provision G.3.d. of WDRs Order R5-2014-0147 is satisfied.

Furthermore, with the discontinued use of Pond 1 through Pond 6 as discussed above, the majority of the WWTF treatment components fall within a Priority 2 Zone. As such, we have re-designated your facility to solely the Priority 2 Merced Sub-basin. The Priority 1 Notice to Comply for the Turlock Sub-basin will be rescinded.